PRECALCULUS · UNIT 4: TRIGONOMETRIC FUNCTIONS

Daylight and Seasons

Model hours of daylight and temperatures with sinusoidal functions.

CP2

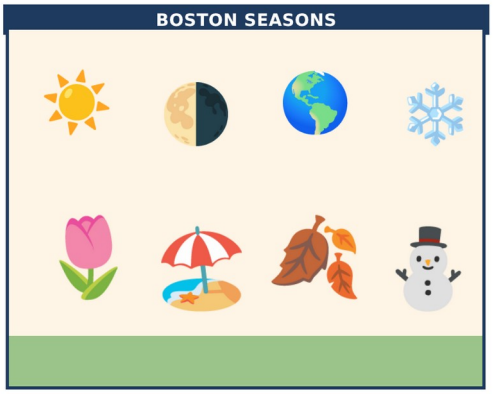
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The parks department wants to schedule outdoor events when there is plenty of daylight and comfortable weather. Daylight and temperature rise and fall every year in a repeating pattern.

You will use the unit circle and sinusoidal functions to model the seasons and recommend event dates.



Longest Day
June 21: 15.3 h

Shortest Day
Dec 21: 9.1 h

Learning Goals

- I can convert between degrees and radians and use the unit circle.
- I can find amplitude, midline, and period from data.
- I can use a sinusoidal model to make predictions.

Project Timeline

When	What you will do
Day 1	Earth's Orbit
Day 2	Daylight Model
Day 3	Plan Events
Final	Seasonal Planning Report + reflection

Part 1: Earth's Orbit

Day 1 · Earth travels around the Sun once every 365 days.

Boston Data Cards

Max Daylight	Min Daylight	Period
15.3 h	9.1 h	365 days
day 172 (June 21)	day 355 (Dec 21)	

Avg Temp High	Avg Temp Low	Event Rule
74°F in late July	29°F in late Jan	Need at least
(day 205)	(day 22)	13 h of daylight

1. How many degrees does Earth travel in one day? (Round to two decimals.)

2. Convert 90° and 180° to radians.

3. Find $\sin\frac{\pi}{6}$, $\cos\frac{\pi}{3}$, and $\sin\frac{\pi}{2}$ using the unit circle.

4. In which quadrant is an angle of $\frac{5\pi}{4}$? Give its sine and cosine signs.

Part 2: Daylight Model

Day 2 · Use the Boston Data Cards.

5. Find the midline (average of max and min daylight).

Starter: *(max + min) ÷ 2*

6. Find the amplitude.

7. What is the period in days?

8. A model is $D(t) = 3.1 \sin\left(\frac{2\pi}{365}(t - 80)\right) + 12.2$. Find D(80) and D(172).

Part 3: Plan Events

Day 3 · Use your model.

9. Find D(1) (January 1) and D(200) (mid-July).

10. Use a table to estimate the first day with at least 13 hours of daylight.

11. Recommend three event dates and explain using the model.

Final Product: Seasonal Planning Report

Your final product must include:

- Degree-radian conversions and unit-circle values
- Amplitude, midline, and period
- Daylight model with predictions
- Event dates
- Seasonal report

Reflection

R1. Why is a sine function a good model for daylight?

R2. What does the midline represent?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Sinusoidal models are built from data and used correctly to predict and solve for event dates. Goes beyond the minimum with a creative, realistic choice.	Sinusoidal models are built from data and used correctly to predict and solve for event dates.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes models, graphs, solutions, and a clear recommendation. Polished, organized, and visually engaging.	Report includes models, graphs, solutions, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.